



Respiratory Distress

Assessment

Pediatric Pearls:

- Use pediatric dosing of medications or electrical therapy for a pediatric patient < 37 kg and as defined by HandTevy.
- Pediatric hypotension is defined as SBP < 70 + (age in years x 2) mmHg

Signs & Symptoms:

- Shortness of breath
- Pursed lip breathing
- Decreased ability to speak
- Increased respiratory rate and effort
- Wheezing, rhonchi, rales, stridor
- Use of accessory muscles
- Fever, cough
- Tachycardia
- Anxious appearance

Differential:

- Asthma / COPD (Emphysema, Bronchitis)
- Anaphylaxis
- Aspiration
- Pleural effusion
- Pneumonia
- Pulmonary embolus
- Pneumothorax
- Cardiac (MI or CHF)
- Pericardial tamponade
- Hyperventilation
- Inhaled toxin (CO, etc.)
- Croup / Epiglottitis
- Congenital heart disease
- Trauma
- Hydrocarbon ingestion

Clinical Management Options

P	P	P	P	P	P	• Oxygen • Position of comfort • BLS airway management
L	L	L	L	L	L	
1	2	3	4	5	6	• If wheezing (non-cardiac), then assist with patient's MDI 2 puffs PRN -or- Albuterol with Ipratropium • If wheezing, Consider CPAP with PEEP if refractory to NEB • If stridor, 3 ml nebulized Isotonic Crystalloid • If Wheezing or Stridor, nebulized Epinephrine
						• Vascular access
						• Monitoring and interpretation of ECG & EtCO₂ • If wheezing or Stridor, Dexamethasone • Epinephrine • If wheezing, Magnesium Sulfate • Advance airway management as needed • If severe bronchospasm refractory to other medications, Ketamine • Ultrasound pleural cavity if pneumo/hemothorax is suspected • Rapid sequence induction as needed to secure patent airway for oxygenation and ventilation.

Consult Online Medical Control As Needed

Pearls:

- Refer to drug formulary charts for all medication dosing for both adults and pediatric patients.
- EtCO₂ and SpO₂ must be monitored continuously if either are abnormal or decline in patient's mental status/condition.
- A silent chest in respiratory distress is a sign for pre-respiratory arrest.
- Chronic COPD may have elevated CO₂ at baseline. Patient respiratory status must be reassessed after each nebulizer or medication administration to determine need for additional dosing.
- CPAP if continued respiratory distress and if adequate mask seal can be established.
- Immediately assess for pneumothorax in asthmatics who develop a sudden decrease in blood pressure, increase in heart rate, or other signs of pneumothorax during an exacerbation.
- Development of bradycardia in respiratory distress in an ominous sign